Audited Flow-Induced Quotient Flow Matching

Fixed-Axis Endpoint Factorization with Low-Dimensional Iterative Dynamics

Theory, algorithm, and empirical audit draft

23 August 2026

Abstract

A latent flow is efficient because its expensive iterative vector field acts on a low-dimensional state, but the usual variational-autoencoder tokenizer can discard density, spend rate matching a fixed prior, and choose a code for reconstruction rather than for closure of the generative dynamics. A full ambient flow avoids that bottleneck but repeatedly processes the ambient dimension. We give an affirmative result on a declared structural class and a falsifiable algorithm outside that class. Flow-Induced Quotient Flow Matching (FIQ-FM) learns a static orthogonal active subspace directly from ordinary flow-matching labels, groups already-fixed residual axes by a cross-fitted conditional-dependence graph, integrates a deep flow only in the active coordinates, and samples a one-shot stochastic block fiber for the complementary randomness. The graph does not recover a generic residual rotation. For triangular transports with block-local Gaussian fibers, FIQ-FM reproduces the exact full-dimensional endpoint law and has iterative complexity governed by the active dimension. Against a baseline restricted to the same chart, active marginal, conditioning, and mean but a diagonal Gaussian residual family, it has an explicit conditional KL advantage. These are representation results for declared model classes, not a proof that the algorithm discovers that class from arbitrary data or dominates an unrestricted VAE. The analysis also gives raw-label active-subspace recovery, exact loss isometry, exact density-error decomposition, pair-partition recovery under a matching margin, and a Wasserstein perturbation bound when the learned field is Lipschitz. The empirical evidence is under regeneration: archived five-seed tables lack their referenced seed-level files, and the original digits command failed during aggregation. The result is conditional rather than universal: arbitrary full flows need not admit cheap quotients, deterministic $d < D$ decoders cannot match full-dimensional densities, and the present dense chart does not establish image-scale speedups.

1 Problem, impossibility boundary, and approach registry

Let $X_1 \sim P_1$ be data in $\mathbb{R}^D$, let $X_0 \sim \mathcal{N}(0, I_D)$ be independent source noise, and let

$$X_t = (1-t)X_0 + tX_1, \qquad V = X_1 - X_0, \qquad t \in [0,1]. \tag{1}$$

The population rectified-flow vector field is

$$v_t^*(x) = \mathbb{E}[V \mid X_t = x]. \tag{2}$$

A full flow estimates $v_t^* : \mathbb{R}^D \to \mathbb{R}^D$. A latent flow estimates a field on $\mathbb{R}^d$, $d \ll D$, but must also specify how the discarded coordinates are generated.

1.1 Why deterministic compression cannot give the requested theorem

Proposition 1.1 (Support obstruction). *Let $d < D$, let Z have any probability law on $\mathbb{R}^d$, and let $G : \mathbb{R}^d \to \mathbb{R}^D$ be locally Lipschitz. Then $G_\#\mathcal{L}(Z)$ is supported on a countably d-rectifiable set and is singular with respect to D-dimensional Lebesgue measure. Consequently, it cannot equal a target law with a nonzero ambient density.*

Proof. A locally Lipschitz image of $\mathbb{R}^d$ is countably d-rectifiable. Every such set has zero D-dimensional Lebesgue measure when $d < D$. The pushforward assigns probability one to that null set, whereas an absolutely continuous target assigns probability zero to it. $\square$

The complementary randomness is therefore essential. A valid full-density factorization must retain a fiber variable $R \in \mathbb{R}^{D-d}$ rather than replace it by a deterministic reconstruction.

1.2 Approach families and selection

The audit began with incompatible routes. The durable conclusions are:

1. **Deterministic autoencoding** is ruled out for full-dimensional targets by the support obstruction.
2. **Train a full flow and distill it** can inherit quality but pays the full training cost first, so it does not solve the stated efficiency problem.
3. **Bottleneck only the ambient denoiser** fails when normal or residual dynamics depend on the full state.
4. **Tangent-only manifold flows** ignore off-manifold density and therefore do not cover noisy data.
5. **A fully moving reversible chart** gives the broadest conjugacy theorem, but without gauge and compute constraints the chart can absorb the entire generator.
6. **A static orthogonal quotient plus a stochastic cheap fiber** has an exact source law, an exact squared-loss isometry, cannot hide a nonlinear generator, and gives the smallest complete theorem.

The selected first implementation is therefore deliberately restricted. The broader moving-chart formulation remains useful as an extension, but its image-scale computational claim is not treated as proved here.

2 Flow-Induced Quotient Flow Matching

The construction separates an expensive active transport from a stochastic residual fiber while keeping an invertible ambient chart.

2.1 Orthogonal quotient and induced field

Let $Q = [Q_z, Q_r] \in \mathbb{R}^{D\times D}$ be orthogonal, with $Q_z \in \mathbb{R}^{D\times d}$. Define

$$Y = Q^\top X = (Z, R), \quad Z = Q_z^\top X, \quad R = Q_r^\top X. \tag{3}$$

Because $Q^\top X_0 \sim \mathcal{N}(0, I_D)$, the source remains factorized standard Gaussian. A structured coordinate field is

$$a_t(z, r) = \begin{bmatrix} u_t(z) \\ w_t(r, z) \end{bmatrix}, \tag{4}$$

where the expensive nonlinear iterative field u_t does not depend on r, while w_t belongs to a cheap family. The induced ambient field is

$$f_t(x) = Q\, a_t(Q^\top x). \tag{5}$$

Theorem 2.1 (Exact source and loss isometry)**.** *For every orthogonal Q and every coordinate predictor a_t,*

$$\left\| V - Q a_t(Q^\top X_t) \right\|_2^2 = \left\| Q^\top V - a_t(Q^\top X_t) \right\|_2^2 \quad \textit{pointwise.} \tag{6}$$

Moreover $Q^\top X_0 \sim \mathcal{N}(0, I_D)$.

Proof. Orthogonality gives $\|b\|_2 = \left\|Q^\top b\right\|_2$ for every b. Apply this to $b = V - Q a_t(Q^\top X_t)$ and use $Q^\top Q = I_D$. Rotational invariance gives the source statement. □

Thus the quotient can be trained directly from ordinary conditional flow labels: no full ambient teacher network is needed.

2.2 Endpoint fiber generator

The audited implementation uses a deep ODE only for Z and a one-shot conditional fiber at the endpoint:

$$U \sim \mathcal{N}(0, I_d), \qquad Z = F(U, C), \tag{7}$$

$$E \sim \mathcal{N}(0, I_{D-d}), \qquad R = \mu(Z, C) + L(Z, C)E, \tag{8}$$

$$X = Q_z Z + Q_r R. \tag{9}$$

Here C is optional conditioning information, F is represented by the latent flow, and L is block lower-triangular with small blocks. The fiber is stochastic, so the endpoint distribution can remain full dimensional.

Definition 2.2 (Exact triangular quotient class)**.** *A conditional target $P_1(\cdot \mid C)$ belongs to $\mathfrak{T}(d, b)$ if there are an orthogonal Q, a diffeomorphism $F(\cdot, C) : \mathbb{R}^d \to \mathbb{R}^d$, and functions μ, L such that (7)–(9) generate $P_1(\cdot \mid C)$, where every block of L has size at most b and is nonsingular.*

Theorem 2.3 (Exact reproduction on the declared factorization class)**.** *Suppose $P_1(\cdot \mid C) \in \mathfrak{T}(d, b)$. If the latent flow exactly transports $\mathcal{N}(0, I_d)$ to $\mathcal{L}(Z \mid C)$, the fiber model exactly represents μ, L, and the ODE is integrated exactly, then FIQ-FM generates exactly $P_1(\cdot \mid C)$.*

Proof. The latent flow produces a random variable with the same conditional law as Z. Conditional on this Z and C, (8) has exactly the target conditional Gaussian law of R. Therefore (Z, R, C) has the target joint law by disintegration. The orthogonal map $(z, r) \mapsto Q_z z + Q_r r$ is bijective, so its pushforward is exactly the target law of X. □

This representational theorem gives the precise sense in which generation quality can equal a full ambient flow while iteration occurs only in dimension d. Membership in $\mathfrak{T}(d, b)$ is assumed rather than discovered, and the theorem does not assert that every data distribution belongs to this class.

3 Exact density decomposition and fixed-chart residual separation

The invertible chart exposes separate active and conditional-fiber errors, which permits a precise comparison to a restricted diagonal residual family.

3.1 Density and KL chain rule

Because Q is volume preserving,

$$q_X(x \mid c) = q_Z(Q_z^\top x \mid c) q_{R|Z}(Q_r^\top x \mid Q_z^\top x, c). \tag{10}$$

Theorem 3.1 (Exact full-density error decomposition)**.** *Let P and Q be two conditional laws on X using the same orthogonal chart. Whenever the KL divergences exist,*

$$\begin{aligned} \mathrm{KL}(P_X(\cdot \mid C) \| Q_X(\cdot \mid C)) &= \mathrm{KL}(P_Z(\cdot \mid C) \| Q_Z(\cdot \mid C)) \\ &\quad + \mathbb{E}_{P_Z}\left[\mathrm{KL}(P_{R|Z}(\cdot \mid Z, C) \| Q_{R|Z}(\cdot \mid Z, C))\right]. \end{aligned} \tag{11}$$

Proof. The orthogonal change of variables has unit absolute determinant. Apply the probability chain rule to $p(z, r \mid c) = p(z \mid c) p(r \mid z, c)$ and take the P expectation of the log density ratio. □

There is no variational gap in (11); latent and fiber errors are explicit and separately measurable.

3.2 Diagonal-decoder gap

Proposition 3.2 (Optimal diagonal Gaussian gap)**.** *Fix (z, c) and suppose the true conditional fiber is $\mathcal{N}(\mu, \Sigma)$ with $\Sigma \succ 0$. Among Gaussian approximations with the correct mean and diagonal covariance D, the KL minimizer is $D = \mathrm{diag}(\Sigma)$ and the minimum is*

$$\Delta_{\mathrm{diag}}(z, c) = \frac{1}{2} \log \frac{\det \mathrm{diag}(\Sigma(z, c))}{\det \Sigma(z, c)} \geq 0. \tag{12}$$

It is strictly positive exactly when Σ is not diagonal.

Proof. For Gaussians with equal mean,

$$\mathrm{KL}(\mathcal{N}(\mu, \Sigma) \| \mathcal{N}(\mu, D)) = \frac{1}{2}\{\mathrm{tr}(D^{-1}\Sigma) - (D - d_r) + \log\det D - \log\det\Sigma\}.$$

Minimizing coordinatewise gives $D_{jj} = \Sigma_{jj}$. Substitution yields (12); Hadamard's inequality gives nonnegativity and its equality condition. □

Corollary 3.3 (Fixed-chart diagonal-residual separation)**.** *Fix the same chart Q, active variable Z, conditioning C, active marginal, and conditional residual mean for both methods. If the baseline residual family is diagonal Gaussian, then for a target in $\mathfrak{T}(d, b)$ with positive expected gap $\mathbb{E}\Delta_{\mathrm{diag}}(Z, C) > 0$, an exact block fiber has zero conditional fiber KL while every such diagonal residual model has full-density KL at least $\mathbb{E}\Delta_{\mathrm{diag}}(Z, C)$.*

This is a fixed-representation model-class separation, not a theorem that the learned latent representation is universally better than a VAE. A baseline allowed to change the chart or use an unrestricted stochastic decoder can remove the gap.

3.3 Rate distortion for KL regularization

Let a VAE encoder define $q(z \mid x)$ and let $p_0(z)$ be a fixed prior. Its average KL obeys

$$\mathbb{E}_X \operatorname{KL}(q(Z \mid X) \| p_0) = I_q(X; Z) + \operatorname{KL}(q_Z \| p_0) \geq I_q(X; Z). \tag{13}$$

For any distortion $d(x, \hat{x})$, rate-distortion theory implies

$$\mathbb{E} d(X, \widehat{X}) \geq D_P(I_q(X; Z)), \tag{14}$$

where D_P is the distortion-rate function. FIQ-FM does not force the aggregated active law toward a fixed prior at encoding time; it learns that law by a flow and retains complementary randomness explicitly. This provides a second conditional separation, although it does not imply dominance over an unrestricted stochastic autoencoder.

4 Learning the quotient from raw flow labels

The first identification result treats a static linear chart; the residual-basis problem is then separated from active-subspace recovery.

4.1 Population moment identity

Assume for this subsection that $\mathbb{E} X_1 = 0$, $\operatorname{Cov}(X_1) = \Sigma$, and $t \in (0, 1)$ is fixed. From (1),

$$\begin{aligned} \mathbb{E}[V X_t^\top] &= \mathbb{E}[(X_1 - X_0)((1-t)X_0 + tX_1)^\top] \\ &= t\Sigma - (1-t) I_D. \end{aligned} \tag{15}$$

Thus the raw flow-label moment has exactly the same eigenvectors as the data covariance.

Theorem 4.1 (Active-subspace recovery)**.** *Suppose*

$$\Sigma = Q_z \Lambda_z Q_z^\top + \lambda_r Q_r Q_r^\top, \quad \min_j (\Lambda_{z,jj} - \lambda_r) = \delta > 0. \tag{16}$$

Then the leading d eigenspace of $M_t = \mathbb{E}\operatorname{sym}(V X_t^\top)$ is $\operatorname{span}(Q_z)$, with eigengap $t\delta$. If $\left\| \widehat{M}_t - M_t \right\|_{\mathrm{op}} \leq \eta < t\delta/2$, then

$$\left\| \sin \Theta(\widehat{Q}_z, Q_z) \right\|_{\mathrm{op}} \leq \frac{2\eta}{t\delta}. \tag{17}$$

Proof. Equation (15) is a scalar affine transformation of Σ, so it preserves eigenvectors and multiplies covariance eigengaps by t. Apply the Davis–Kahan sine theorem. $\square$

For sub-Gaussian samples, matrix Bernstein or standard covariance concentration gives $\eta = O_{\mathbb{P}}(K^2 \sqrt{(D + \log(1/\alpha))/n})$ up to the usual lower-order term. The moment needs one sampled path time and one outer product, or a randomized sketch, per observation; no ambient neural vector field is fitted.

4.2 The unresolved residual gauge

The active theorem does *not* identify the basis of $\mathrm{span}(Q_r)$ when the residual eigenvalues are tied. If $O \in \mathbb{R}^{(D-d)\times(D-d)}$ is orthogonal, then $[Q_z, Q_r O]$ recovers the same active quotient. A cheap block covariance in one residual basis can become dense after rotation by O.

The present implementation does not solve this rotation problem. It estimates conditional-dependence scores after a residual basis has been fixed and only permutes those axes into blocks. The exact benchmark has unequal residual marginal variances, so its covariance eigenvectors already choose residual axes; it cannot validate recovery under a generic residual rotation.

4.3 Cross-fitted conditional covariance graph

After fixing Q_z, define provisional residuals $R = Q_r^\top X$. Centering can be included; in the exact benchmark the conditional means are zero. For $i \neq j$, define the population edge weight

$$S_{ij} = \mathrm{Var}(\mathbb{E}[R_i R_j \mid Z, C])\,. \tag{18}$$

The implementation estimates $\mathbb{E}[R_i R_j \mid Z, C]$ on the training split using a fixed nonlinear feature dictionary and evaluates predictive R^2 on validation data. Test data are never used for the graph.

Assumption 4.2 (Pair-block margin)**.** *The residual coordinates admit a true partition* $\mathcal{P}^* = \{\{i_1, j_1\}, \ldots, \{i_m, j_m\}\}$ *such that*

$$\sum_{\{i,j\}\in\mathcal{P}^*} S_{ij} \geq \sum_{\{i,j\}\in\mathcal{P}} S_{ij} + \gamma \tag{19}$$

for every other perfect matching $\mathcal{P}$*, for some* $\gamma > 0$.

Theorem 4.3 (Pair-partition recovery on fixed residual axes)**.** *Let* $\widehat{S}$ *satisfy* $\max_{i<j} |\widehat{S}_{ij} - S_{ij}| \leq \varepsilon$. *If the pair-block margin holds and* $2m\varepsilon < \gamma$*, then maximum-weight perfect matching under* $\widehat{S}$ *recovers* $\mathcal{P}^*$.

Proof. For any matching $\mathcal{P}$, its empirical total differs from its population total by at most $m\varepsilon$. Hence the empirical gap between $\mathcal{P}^*$ and any competitor is at least $\gamma - 2m\varepsilon > 0$. □

The theorem recovers a partition, not an orthogonal residual basis. For blocks larger than pairs, the code uses a greedy score grouping. That is a practical heuristic, not covered by this result.

5 Algorithm

Stage A: quotient discovery.

1. Split data into train, validation, and test before model fitting.
2. Sample raw rectified-flow triples (T, X_T, V) from train data and Gaussian source noise.
3. Estimate $\widehat{M} = \frac{1}{n}\sum_i \mathrm{sym}(V_i X_{T_i,i}^\top)$ and take its leading d eigenvectors as $\widehat{Q}_z$.
4. Choose an arbitrary orthogonal residual completion $\widehat{Q}_r$.
5. Estimate validation conditional-dependence weights $\widehat{S}_{ij}$ and permute/group residual axes by maximum-weight pair matching or the declared block heuristic.

Stage B: active flow. Transform each observation to $(Z, R) = \widehat{Q}^\top X$. Train a rectified-flow field $u_\theta(t, z, c)$ from

$$\min_\theta \mathbb{E} \left\| Q_z^\top V - u_\theta(T, Q_z^\top X_T, C) \right\|^2. \tag{20}$$

The population target is the active conditional velocity. In the exact quotient class it closes in (z, c).

Stage C: one-shot fiber. Train block-Cholesky functions (μ_ψ, L_ψ) by

$$\min_\psi -\mathbb{E} \log \mathcal{N}(R; \mu_\psi(Z, C), L_\psi(Z, C) L_\psi(Z, C)^\top). \tag{21}$$

All axes are retained; only covariance interactions are restricted to the declared blocks.

Stage D: generation. Sample U, E independently from standard Gaussian, integrate the d-dimensional active ODE to get Z, sample $R = \mu_\psi(Z, C) + L_\psi(Z, C)E$ once, and decode $X = Q_z Z + Q_r R$.

Stage E: falsification. On held-out validation data, report active closure error, block-versus-diagonal fiber NLL, dimension sweeps, and total inference cost. Increase d or fiber complexity when the approximation error is larger than the predeclared compute penalty; do not claim a quotient when the residual field remains complex.

6 Complexity

Let $C_u(d)$ be one evaluation cost of the active vector field, K the number of ODE evaluations, $C_f(D, d, b)$ the one-shot block-fiber cost, and $C_Q(D)$ the chart cost. Then

$$C_{\mathrm{FIQ}} = K C_u(d) + C_f(D, d, b) + C_Q(D), \tag{22}$$

$$C_{\mathrm{full}} = K C_v(D), \tag{23}$$

$$C_{\mathrm{latent}} = K C_u(d) + C_{\mathrm{decoder}}(D, d). \tag{24}$$

For fixed small b, the block fiber is linear in the number of residual coordinates up to the context network. The current dense orthogonal chart has $C_Q(D) = O(D^2)$ once. Therefore:

- the *iterative* complexity is exactly latent-dimensional;
- total complexity has the same form as latent flow plus a one-shot decoder/fiber;
- a proof of image-scale latent-order complexity additionally requires a structured chart with $C_Q(D) = O(D \log D)$ or $O(D)$, such as local lifting, wavelets, sparse couplings, or butterfly factors.

The experiments count the dense chart and fiber in wall-clock inference time; they do not claim orders-of-magnitude image speedups.

7 Finite-sample and approximation bounds

The following results separate estimation error in the active field from endpoint error and model-selection uncertainty.

7.1 Active field oracle inequality

Let $\mathcal{U}_d$ be a bounded active-field class, and let $B_z = Q_z^\top V$. Define the approximation error

$$A_d^2 = \inf_{u \in \mathcal{U}_d} \mathbb{E} \left\| B_z - u(T, Z, C) \right\|^2 - \mathbb{E} \left\| B_z - u_d^*(T, Z, C) \right\|^2, \tag{25}$$

where $u_d^* = \mathbb{E}[B_z \mid T, Z, C]$ is unrestricted. For empirical risk minimization $\widehat{u}_d$ under bounded squared loss, a standard Rademacher argument gives, with probability at least $1 - \alpha$,

$$\mathbb{E} \left\| \widehat{u}_d - u_d^* \right\|^2 \lesssim A_d^2 + \mathfrak{R}_n(\ell \circ \mathcal{U}_d) + B^2 \sqrt{\frac{\log(1/\alpha)}{n}}. \tag{26}$$

The scientific content is not the generic empirical-process inequality; it is that A_d is the observable closure boundary. If A_d remains large, the claimed quotient is false at that dimension.

7.2 Endpoint perturbation

Let $Y_t = (Z_t, R_t)$ follow the exact structured coordinate field a_t^* and let $\widehat{Y}_t$ follow $\widehat{a}_t$. Suppose both use the same source, $\widehat{a}_t$ is L-Lipschitz uniformly in time, both solutions have finite second moments, and

$$\int_0^1 \mathbb{E} \left\| \widehat{a}_t(Y_t) - a_t^*(Y_t) \right\|^2 dt \le \varepsilon^2. \tag{27}$$

Proposition 7.1 (Wasserstein stability)**.** *Under the preceding assumptions and exact integration,*

$$W_2(\mathcal{L}(\widehat{X}_1), \mathcal{L}(X_1)) \le e^L \varepsilon, \tag{28}$$

where $X = QY$.

Proof. Couple the solutions through the same source and write $\Delta_t = \widehat{Y}_t - Y_t$. The identity

$$\dot{\Delta}_t = \widehat{a}_t(\widehat{Y}_t) - \widehat{a}_t(Y_t) + \widehat{a}_t(Y_t) - a_t^*(Y_t)$$

uses the Lipschitz assumption on $\widehat{a}_t$. Grönwall and Cauchy–Schwarz in time give $\{\mathbb{E} \left\| \Delta_1 \right\|^2\}^{1/2} \le e^L \varepsilon$. Orthogonality preserves Euclidean distance, and this coupling upper bounds W_2. $\square$

Equation (28) uses regularity of $\widehat{a}_t$ and the true-path error in (27). A bound estimated under another path additionally needs a change-of-measure control.

7.3 Dimension and block selection

Let $\widehat{R}_d$ be held-out active-plus-fiber loss, let $\mathrm{pen}_n(d, b)$ be a capacity penalty, and let $\mathrm{cost}(d, b)$ include measured FLOPs or time. Select

$$(\widehat{d}, \widehat{b}) = \mathrm{argmin}_{d,b} \left\{ \widehat{R}_d(b) + \mathrm{pen}_n(d, b) + \lambda \, \mathrm{cost}(d, b) \right\}. \tag{29}$$

A finite candidate-set union bound yields an oracle inequality against the best candidate up to twice the uniform validation deviation. This formalizes the bias–variance–compute tradeoff; it does not manufacture a low-dimensional quotient when no candidate has small held-out defect.

8 Experimental protocol

The protocol fixes information, held-out use, metrics, and cost accounting before any archived or fresh result is interpreted.

8.1 Fairness and leakage controls

Every seed is split before fitting. Standardization, chart estimation, fixed-axis residual partitioning, early stopping, VAE/RAE encoders, decoders, linear-probe tuning, and evaluator selection use only train and validation data. Test data are used once for final metrics. All generative methods receive the same examples, labels, active dimension, sample count, and Heun NFE. FIQ, VAE-LFM, and RAE-LFM use the same active vector-field architecture. The full ambient width is chosen so its trainable generation parameters do not exceed the FIQ generation budget. Endpoint decoder/fiber and chart time are included.

The five recorded distribution metrics are sliced W_2, an adaptive-bandwidth Gaussian-kernel MMD U-statistic clipped at zero, energy distance, covariance error, and normalized mean error. Digits additionally use a train-only feature/classifier network to report feature Fréchet, class-conditional feature Fréchet, feature-space precision/recall, and requested-label accuracy. Because the old digits seeds reuse the same finite dataset, their resampling intervals describe split sensitivity rather than independent scientific replication.

8.2 Archived exact-quotient snapshot

The following table preserves a pre-audit snapshot. The referenced seed-level files are absent, the checked summary schema was not emitted by the current script, and the old evaluator used method-dependent randomness. The values are not promoted until a fresh registered run regenerates them from complete raw records.

The target has $D = 18$, $d = 2$, a nonlinear active diffeomorphism, and eight conditionally correlated 2×2 Gaussian residual blocks, hidden by a random orthogonal chart. Each seed uses 5,000 train, 1,500 validation, and 2,500 test examples. The baselines are:

1. FIQ-FM with block fiber;
2. the same model with diagonal fiber;
3. a parameter-matched full ambient flow;
4. KL-VAE latent flow with diagonal stochastic decoder;
5. representation-autoencoder latent flow with block stochastic decoder.

Table 1: Archived exact quotient snapshot, five seeds. Lower is better. Mean $\pm$ standard error.

Method	sliced W_2	MMD^2	energy	covariance error
FIQ-FM block	$\mathbf{0.0400 \pm 0.0016}$	$\mathbf{0.00044 \pm 0.00020}$	$\mathbf{0.00245 \pm 0.00063}$	$\mathbf{0.0433 \pm 0.0026}$
FIQ-FM diagonal	0.0642 ± 0.0029	0.00158 ± 0.00053	0.00769 ± 0.00189	0.1119 ± 0.0048
Full FM matched	0.0614 ± 0.0016	0.00282 ± 0.00055	0.01265 ± 0.00174	0.0596 ± 0.0061
KL-VAE LFM	0.0708 ± 0.0016	0.00150 ± 0.00031	0.00889 ± 0.00115	0.1203 ± 0.0047
RAE LFM block	0.0638 ± 0.0013	0.00094 ± 0.00037	0.00575 ± 0.00131	0.0995 ± 0.0080

The archived report recorded a paired sliced-W_2 improvement of 0.0242 versus the diagonal ablation, 0.0214 versus the finite-recipe full FM, 0.0307 versus KL-VAE, and 0.0237 versus RAE. It also recorded an active-subspace sine error of 0.020, a fitted diagonal-minus-block NLL gain of 1.292 nats, an analytic gap of 1.291 nats, and median CPU generation 1.30× faster than the matched full flow. These are provenance-incomplete historical values, not current empirical evidence.

8.3 Archived real-digits snapshot under misspecification

The following table preserves a pre-audit snapshot. Its referenced seed-level files are absent, and the original aggregation command failed. The values are not promoted until a fresh registered run regenerates them from complete seed-level records.

The real-data experiment uses the 1,797 sklearn 8×8 digit images. Each seed uses a stratified 60/20/20 split, train-only standardization, fixed split-specific uniform dequantization, active dimension $d = 16$, 20 Heun steps, and equal label information.

Table 2: Archived sklearn digits snapshot, five seeds. Higher accuracy and lower distances are better.

Method	class feature Fréchet	sliced W_2	label accuracy	feature Fréchet
FIQ-FM block	$\mathbf{11.002 \pm 0.646}$	$\mathbf{0.307 \pm 0.006}$	$\mathbf{0.658 \pm 0.015}$	$\mathbf{5.239 \pm 0.484}$
FIQ-FM diagonal	11.289 ± 0.800	0.318 ± 0.005	0.628 ± 0.013	5.327 ± 0.606
Full FM matched	33.820 ± 1.359	0.286 ± 0.002	0.342 ± 0.015	18.922 ± 0.822
KL-VAE LFM	11.990 ± 0.850	0.345 ± 0.006	0.607 ± 0.014	6.179 ± 0.678
RAE LFM block	11.131 ± 0.791	0.337 ± 0.005	0.629 ± 0.015	5.342 ± 0.636

The archived table reported a sliced-W_2 advantage of 0.0384 over KL-VAE and 0.0305 over RAE. It reported requested-label accuracy advantages of 0.0506 and 0.0294, respectively. Because the raw seed files are missing and the resamples overlap, neither the old intervals nor their significance labels are treated as current evidence.

The archived table also recorded conditional-mean reconstruction MSEs of 0.190 ± 0.005 for FIQ, 0.420 ± 0.003 for KL-VAE, and 0.373 ± 0.003 for RAE, with linear-probe accuracies 0.961, 0.954, and 0.965. These descriptive values point in different directions and do not establish that one representation is better on every axis.

9 Failure trace and adversarial audit

1. **Residual-basis boundary:** the active theorem leaves generic rotations inside a tied residual eigenspace unresolved. The implemented graph only groups axes after that basis is fixed.

2. **Benchmark limitation:** unequal residual marginal variances select the synthetic residual axes before grouping, so this benchmark tests pair partitioning but not residual-rotation recovery.

3. **Support check:** every deterministic $d < D$ proposal was rejected for full-density parity.

4. **Compute check:** the chart and fiber are included in timing. The dense chart prevents an image-scale asymptotic claim.

5. **Baseline check:** removing the VAE KL penalty and giving the RAE a block stochastic decoder produces a materially stronger baseline; it is retained.

6. **Metric check:** digits feature Fréchet is not declared significantly better when the paired interval includes zero.

7. **Information check:** the same labels and splits are supplied to every method; no group or test labels enter model selection.

8. **Proof boundary:** exact pair matching is proved; greedy blocks of size four on digits are not.

The detailed registry and audit are stored with the code in `theory/approach_registry.md` and `theory/adversarial_audit.md`.

10 What is and is not established

Established. On $\mathfrak{T}(d, b)$, the algorithm has exact full-dimensional endpoint law under realizability and exact integration, exact latent/fiber KL decomposition, and iterative dynamics only in dimension d. Raw flow labels recover the active subspace in the spiked linear-Gaussian regime. Pair partitions are identifiable under a matching margin. A diagonal stochastic VAE decoder has the explicit strict gap (12). Fresh synthetic and observed-data confirmation remains necessary before making empirical claims.

Not established. There is no universal cheap quotient, no universal dominance over an unrestricted VAE, no proof that the greedy block-four grouping is optimal, and no demonstrated parity with a converged full pixel flow on modern image benchmarks. A structured reversible image chart, convolutional or tokenized block fiber, compute-matched DiT baselines, and image-scale FID/likelihood experiments are required for that next claim.

11 Reproduction

The package is self-contained:

```
python -m pip install -e .
pytest -q
python experiments/synthetic_exact/run.py --seeds 0 1 2 3 4 \
  --output results/synthetic_exact_confirmation
python experiments/sklearn_digits/run.py --seeds 0 1 2 3 4 \
  --output results/sklearn_digits_confirmation
```

Five executable theorem and leakage checks cover orthogonal round-trip and loss isometry, moment subspace recovery, positivity of the analytic KL gap, split disjointness, and recovery of deliberately scrambled residual pairs. Configurations, every seed-level metric, paired summaries, and training records are persisted by the scripts.

12 Conclusion

The mathematically coherent route is not to compress the endpoint and hope that a low-dimensional code happens to be generatively sufficient. It is to identify a quotient of the full generative law, preserve complementary randomness, and spend iterative computation only where the transport is genuinely nonlinear. A correct active subspace can still yield an inefficient fiber in the wrong residual basis. The present algorithm groups fixed axes but does not learn that basis under a generic rotation. Conditional on an aligned block-local stochastic fiber, the model gives exact full-density parity on its declared class and a strict separation from diagonal VAE decoders. Image-scale claims require a structured chart, a small held-out quotient defect, a cheap fiber, residual-basis discovery or a rotation-robust fiber, and compute-matched full-flow quality.